

D553

Data Analytics for Accountants II

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A controller regresses monthly utility cost on machine hours and gets $\text{Cost} = \$4,200 + \$3.50 \times \text{machine hours}$, with $R^2 = 0.88$. The plant was built in 2008 and runs two shifts. Next month the plant expects 6,000 machine hours. What is the predicted utility cost?

- A. \$14,700
- B. \$25,200
- C. \$21,000
- D. \$22,176

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Utility cost follows $\text{Cost} = \$4,200 + \$3.50 \times \text{machine hours}$. Last month's cost was \$18,200. How many machine hours were used?

- A. 14,000
- B. 63,700
- C. 5,200
- D. 4,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A tax analyst models quarterly sales-tax liability using store traffic and gets $R^2 = 0.64$. The model was built in Python by a summer intern. What does 0.64 mean?

- A. Store traffic explains 64% of the variation in sales-tax liability.
- B. The correlation between traffic and liability is 0.64.
- C. The model's predictions are exactly right 64% of the time.
- D. Store traffic causes 64% of the liability.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

A simple regression has a correlation coefficient of $r = -0.9$. What is R^2 ?

- A. 0.90
- B. 0.95
- C. 0.81
- D. -0.81

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An analyst forecasts next month's product returns with a 3-month simple moving average. Returns for the last five months, oldest first, were 400, 460, 520, 480 and 500 units. The company recently switched to a new returns portal. What is the forecast?

- A. 1,500 units
- B. 500 units
- C. 460 units
- D. 472 units

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A 3-month moving-average forecast for July is 600 units. April was 550 units and May was 620 units. How many units were there in June?

- A. 630
- B. 600
- C. 1,800
- D. 585

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Sales at a lawn-care supply company rise every April through June and fall every November through January, year after year. The company opened its fifth warehouse last year. Which time-series component does this pattern show?

- A. Trend, driven by the new warehouse.
- B. Cyclical variation.
- C. Irregular variation.
- D. Seasonality.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

An analyst builds a revenue model with 40 variables that fits the 2019–2023 training data almost perfectly ($R^2 = 0.99$), but it predicts 2024 poorly. The model took three weeks to build on a high-end laptop. What is the most likely problem?

- A. The laptop lacked processing power.
- B. Underfitting; add more variables.
- C. Overfitting; simplify the model and validate it on held-out data.
- D. Nothing; an R^2 of 0.99 always proves the model is right, so 2024 was a fluke.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

In a full-population test of 12,000 travel expense claims, the mean claim is \$840 and the standard deviation is \$160. The audit team flags any claim with a z-score above 3. The claims were filed through a mobile app. Which claims are flagged?

- A. Claims above \$1,000.
- B. Claims above \$360.
- C. Claims above \$1,320.
- D. Claims above \$480.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

A claim of \$1,560 has a z-score of 4.5 in a population with a mean of \$840. What is the standard deviation?

- A. \$720
- B. \$3,240
- C. \$347
- D. \$160

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

Across 60 branches of a bank with a strong regional brand, the correlation between overtime hours and reconciliation errors is $r = +0.72$. All branches use the same accounting software. What is the best interpretation?

- A.** A fairly strong positive association: branches with more overtime tend to have more errors, but this does not prove overtime causes errors.
- B.** Overtime causes 72% of reconciliation errors.
- C.** A perfect relationship, so cutting overtime will eliminate errors.
- D.** A weak negative relationship.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

In a regression predicting audit hours, the coefficient for number of subsidiaries is 12.5 with $p = 0.003$, and the coefficient for client industry code is 40.0 with $p = 0.41$. The audit firm has offices in nine cities. Using a 0.05 level, which conclusion fits?

- A.** Industry code matters more, because its coefficient is larger.
- B.** Number of subsidiaries is a significant predictor; industry code is not, despite its larger coefficient.
- C.** Subsidiaries is not significant, because 0.003 is below 0.05.
- D.** Both are significant, because both coefficients are positive.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A controller regresses monthly utility cost on machine hours and gets $\text{Cost} = \$4,200 + \$3.50 \times \text{machine hours}$, with $R^2 = 0.88$. ~~The plant was built in 2008 and runs two shifts.~~ Next month the plant expects 6,000 machine hours. What is the predicted utility cost?

Target word: predicted cost **Decides:** Plug 6,000 hours into the equation.

A	Inverted operation	Multiplies the intercept by the slope instead of using machine hours.
B	Correct	$\$4,200 + \$3.50 \times 6,000 = \$4,200 + \$21,000 = \$25,200$.
C	Dropped element	Leaves out the \$4,200 fixed intercept.
D	Camouflage	Multiplies the prediction by R^2 , which measures fit, not cost.

2. Answer D Concept 1 • transfer

Utility cost follows $\text{Cost} = \$4,200 + \$3.50 \times \text{machine hours}$. Last month's cost was \$18,200. How many machine hours were used?

Target word: solve for x **Decides:** $(18,200 - 4,200) / 3.50$.

A	Adjacent category	That is the variable cost in dollars, not hours.
B	Inverted operation	Multiplies by the slope instead of dividing.
C	Dropped element	Divides total cost by the slope and ignores the intercept.
D	Correct	$\$14,000 / \$3.50 = 4,000$ hours.

3. Answer A Concept 2

A tax analyst models quarterly sales-tax liability using store traffic and gets $R^2 = 0.64$. ~~The model was built in Python by a summer intern.~~ What does 0.64 mean?

Target word: meaning of R^2 **Decides:** R^2 is explained variation.

A	Correct	R^2 is the share of variance explained by the model.
B	Confusion set	r would be $\sqrt{0.64} = 0.80$.
C	Adjacent category	R^2 is not a hit rate.
D	Absolute language	R^2 describes fit, not cause.

4. Answer C Concept 2 • transfer

A simple regression has a correlation coefficient of $r = -0.9$. What is R^2 ?

Target word: R^2 from r **Decides:** $R^2 = r^2$.

A	Wrong stem number	That is the size of r, not R^2 .
B	Adjacent category	Takes a square root instead of squaring.
C	Correct	$(-0.9)^2 = 0.81$.
D	Inverted operation	R^2 is a square and cannot be negative.

5. Answer B Concept 3

An analyst forecasts next month's product returns with a 3-month simple moving average. Returns for the last five months, oldest first, were 400, 460, 520, 480 and 500 units. ~~The company recently switched to a new returns portal.~~ What is the forecast?

Target word: 3-month moving average **Decides:** Average the most recent three: 520, 480, 500.

A	Dropped element	Sums the three months but does not divide.
B	Correct	$(520 + 480 + 500) / 3 = 500$.
C	Wrong stem number	Averages the oldest three months.
D	Adjacent category	Averages all five months.

6. Answer A Concept 3 • transfer

A 3-month moving-average forecast for July is 600 units. April was 550 units and May was 620 units. How many units were there in June?

Target word: solve for a missing month **Decides:** The three months must total 1,800.

A	Correct	$1,800 - 550 - 620 = 630$.
B	Wrong stem number	That is the forecast, not June.
C	Dropped element	The three-month total; April and May still have to be removed.
D	Adjacent category	Averages April and May instead of solving for June.

7. Answer D Concept 4

Sales at a lawn-care supply company rise every April through June and fall every November through January, year after year. ~~The company opened its fifth warehouse last year.~~ Which time-series component does this pattern show?

Target word: name the component **Decides:** A pattern that repeats within each year.

A	Camouflage	Growth may add a trend, but the repeating yearly pattern is seasonality.
B	Adjacent category	Cycles span several years and follow the business cycle.
C	Confusion set	Irregular movement is random, not a repeating pattern.
D	Correct	Regular within-year peaks and troughs are seasonal.

8. Answer C Concept 5

An analyst builds a revenue model with 40 variables that fits the 2019–2023 training data almost perfectly ($R^2 = 0.99$), but it predicts 2024 poorly. ~~The model took three weeks to build on a high-end laptop.~~ What is the most likely problem?

Target word: diagnose the model **Decides:** Near-perfect in sample, poor out of sample, many variables.

A	Camouflage	Hardware affects speed, not predictive accuracy.
B	Inverted operation	The model is already too complex; more variables make it worse.
C	Correct	The model learned noise in the training data.
D	Absolute language	In-sample fit does not prove predictive power.

9. Answer C Concept 6

In a full-population test of 12,000 travel expense claims, the mean claim is \$840 and the standard deviation is \$160. The audit team flags any claim with a z-score above 3. ~~The claims were filed through a mobile app.~~ Which claims are flagged?

Target word: threshold amount **Decides:** Mean + 3 × SD.

A	Wrong stem number	Uses one standard deviation instead of three.
B	Inverted operation	Subtracts 3 SD from the mean.
C	Correct	$\$840 + 3 \times \$160 = \$1,320$.
D	Dropped element	Drops the mean; \$480 is only 3 × SD.

10. Answer D Concept 6 · transfer

A claim of \$1,560 has a z-score of 4.5 in a population with a mean of \$840. What is the standard deviation?

Target word: solve for SD **Decides:** $SD = (x - \text{mean}) / z$.

A	Adjacent category	That is the distance from the mean, not the SD.
B	Inverted operation	Multiplies the difference by z.
C	Dropped element	Divides the claim by z and ignores the mean.
D	Correct	$(\$1,560 - \$840) / 4.5 = \$720 / 4.5 = \160 .

11. Answer A Concept 7

Across 60 branches of a bank ~~with a strong regional brand~~, the correlation between overtime hours and reconciliation errors is $r = +0.72$. All branches use the same accounting software. What is the best interpretation?

Target word: meaning of r **Decides:** Positive, fairly strong, observational.

A	Correct	Sign and size describe the link; causation needs more evidence.
B	Confusion set	r is not a percentage of cause.
C	Absolute language	Only $r = 1$ is perfect, and correlation is not causation.
D	Inverted operation	+0.72 is positive and fairly strong.

12. Answer B Concept 8

In a regression predicting audit hours, the coefficient for number of subsidiaries is 12.5 with $p = 0.003$, and the coefficient for client industry code is 40.0 with $p = 0.41$. ~~The audit firm has offices in nine cities.~~ Using a 0.05 level, which conclusion fits?

Target word: which predictor is significant **Decides:** $p = 0.003 < 0.05$; $p = 0.41 > 0.05$.

A	Confusion set	A large coefficient with $p = 0.41$ may be noise.
B	Correct	Significance comes from the p-value, not the coefficient's size.
C	Inverted operation	Below the threshold means significant.
D	Dropped element	Ignores the p-values entirely.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	4	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Using a regression equation	Predicted $y = \text{intercept} + \text{slope} \times x$; to solve for x , subtract the intercept and divide by the slope.	Dropping the intercept or scaling the result by R^2 .	An equation of the form $y = a + b \times x$ and a planned x .
2	What R^2 means	R^2 is the share of variation in y explained by the model; in simple regression it equals r^2 .	Reading R^2 as accuracy, as causation, or as the correlation itself.	An R^2 value and a question about what it means.
3	Moving-average forecasts	An n -period simple moving average forecasts the next period as the mean of the most recent n periods.	Averaging all periods or the oldest ones.	A series of past values and a 3-month (or n -month) average.
4	Time-series components	Trend is long-run direction; seasonality repeats within a year; cyclical swings span multiple years; irregular is random.	Calling a yearly repeating pattern a trend or a cycle.	The same months rise and fall every year.
5	Overfitting	A model that fits training data almost perfectly but predicts new data poorly has learned noise; simplify and validate on held-out data.	Trusting a very high training R^2 as proof.	Great fit in sample, poor fit out of sample, many variables.
6	Outliers by z-score	$z = (x - \text{mean}) / \text{standard deviation}$; a threshold of z translates to $\text{mean} + z \times \text{SD}$.	Forgetting the mean or using one standard deviation.	A mean, a standard deviation, and a flag threshold in SDs.
7	Interpreting correlation	r runs from -1 to $+1$; sign shows direction, size shows strength; correlation does not show causation.	Reading r as a percentage caused, or as proof of cause.	An r value between two business measures.
8	Significance of regression coefficients	A coefficient with p below the chosen level (often 0.05) is statistically significant; the coefficient's size alone does not show significance.	Picking the predictor with the biggest coefficient as most important.	Several coefficients listed with p -values.

Course scope (public description): Data Analytics for Accountants II equips students with the skills needed to analyze and present data to make reliable forecasts and propose strategies. Accountants are no longer limited to being the chroniclers of an organization's profits and expenses; they now leverage the power of data to predict the future and advise leaders on appropriate actions. This course will emphasize auditing, managerial accounting, tax accounting and financial reporting as the learner completes a professional scenario that requires predictive analysis and presenting a course of action. As part of the Master of Accounting program, this course will cover content that appears in the data analysis portions of the Certified Public Accountant (CPA) and Certified Management Accountant (CMA) exams. Data Analytics for Accountants I is a prerequisite for this course.